

ASSIGNMENT NO 2

Deep Learning Course

Preface. Deep Learning with Tensorflow & Keras (3rd Edition) has been uploaded on Github Repo for students reference. In Chapter 6 of the Book at page 224 there is an implementation of the Transformer Architecture (from scratch) using Tensorflow and Keras. This architecture was discussed in the class on 28 May 2026 and its implementation was also demonstrated. The Transformer Architecture is a continuation of the Attention Mechanism introduced for RNNs as a sequel to Sequence-to-Sequence Encoder-Decoder Architecture that is mentioned in Chapter 5 of the Book at page 172. This was also explained and discussed in two classes. Read through Sequence to Sequence, Attention Mechanism and Transformers to answer and implement following:-

Requirements

1. What is the difference between Sequence to Sequence Architecture and Attention Mechanism?
2. Explain Transformer Networks and how attention Mechanism is implemented in the Transformers.
3. Implement Transformer Networks from scratch as per the example given in the book at page 224 and share following results.
 - a. Training Loss
 - b. Validation Loss
 - c. The size of Matrices Q, K and V
 - d. Test & Evaluation Results
 - e. Prediction

Use the same repository as mentioned in the Book Example i-e Portugese to English Repo

Useful Links to understand the internal working of a transformer are

1. <https://www.youtube.com/watch?v=eMlx5fFNoYc>
2. <https://www.youtube.com/watch?v=wjZofJX0v4M>

GOOD LUCK